

## Code-Layout, Readability and Reusability

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 15 March 2024                         |
| <b>Team ID</b>       | XXXXXX                                |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 5 Marks                               |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                                                                            |
|------|---------------------------|-----------------------------------------------|-------------------------------|------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           | Proper 4-space indentation followed in all Python files                            |
| 2    | Proper File Structure     | Files and folders are logically organize      | Yes                           | Project divided into data, src, templates, static, models, notebooks               |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Variables such as life_expectancy, education_years, gni_per_capita are descriptive |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Functions like preprocess_data(), train_model(), predict_hdi()                     |
| 5    | Code Comments             | Inline and block comments explain log         | Yes                           | Important functions and ML workflow are documented                                 |
| 6    | Modular Design            | Code is split into reusable functions/m       | Yes                           | Separate files for preprocessing, training, prediction and Flask app               |

|   |                   |                                              |     |                                                               |
|---|-------------------|----------------------------------------------|-----|---------------------------------------------------------------|
| 7 | No Redundant Code | Duplicate or unused code is removed          | Yes | Reusable functions minimize duplication                       |
| 8 | Error Handling    | Exceptions and errors are handled gracefully | Yes | Flask input validation and Python exception handling included |

#### Reusable Components / Modules:

| S.No | Component / Module Name   | Language / Technology         | Where Reused          | Reusability Level (High / Medium / Low) |
|------|---------------------------|-------------------------------|-----------------------|-----------------------------------------|
| 1    | Data Preprocessing Module | Python (Pandas, Scikit-learn) | Training & Prediction | High                                    |

|   |                        |              |                          |        |
|---|------------------------|--------------|--------------------------|--------|
| 2 | Feature Scaling Module | Scikit-learn | Model Training & Testing | High   |
| 3 | Machine Learning Model | Scikit-learn | Flask Prediction System  | High   |
| 4 | Prediction Function    | Python       | Flask Backend            | High   |
| 5 | HTML User Interface    | HTML, CSS    | Web Application          | Medium |

#### Overall Code Quality Assessment:

| Aspect                   | Rating (1-5) | Comments                                                                 |
|--------------------------|--------------|--------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | Well-organized project folders with modular architecture                 |
| Readability              | 5            | Easy-to-understand code with meaningful naming conventions               |
| Reusability              | 5            | Modules can be reused in other ML prediction projects                    |
| Documentation / Comments | 5            | Adequate comments and documentation throughout the project               |
| Overall Score            | 5/5          | Project follows good coding standards and software engineering practices |